

Air-Gap Dependencies: What to Carry, Versions, and Dependency Management

Everything the platform needs on the far side of the air gap - host software, container images, model weights, offline build bundles and code - with the exact versions inside every image, and how those versions are controlled and updated.

Versions are read from the built images themselves (docs/airgap/collect_inventory.py, collected 2026-09-27 16:23), not from requirement files: requirement files say what was asked for, the images show what was actually installed. The same script writes docs/airgap/AIRGAP_BILL_OF_MATERIALS.csv (one row per shipped component) - run it again before every transfer, and inside the gap to prove the images arrived unchanged. This document supersedes docs/air_gapped_dep.md, which still describes CUDA 12.1, PyTorch 2.2 and a 4 GB image.

Contents

1. What to carry - the checklist
2. What each node needs
3. Platform versions per image
4. Python packages and why they are pinned
5. System packages (.deb bundle)
6. Model weights
7. Version compatibility rules
8. Dependency management
9. Packaging, transfer and verification
10. Risks found in this inventory
11. Appendix A - every Python package per image
12. Appendix B - .deb bundle
13. Appendix C - model file checksums

1. What to carry - the checklist

#	item	form	size	needed on	changes when
1	Host software: Docker Engine + Compose v2 plugin, NVIDIA driver, NVIDIA Container Toolkit	OS packages / installers (installed before the gap, or carried as .deb / .rpm / .run)	varies	every node (driver + toolkit: GPU nodes)	rarely
2	spark-lean image	docker save .tar.gz	0.88 GB	Spark nodes, service mode	Spark / Python change
3	ner-translate-server image (the kitchen)	docker save .tar.gz	3.26 GB	GPU model-server node	any kitchen dependency change
4	ner-translate-worker image	docker save .tar.gz	10.59 GB	Spark nodes, cluster mode	NER dependency change (or rebuild inside, row 8)
5	multi-model-inference image	docker save .tar.gz	10.47 GB	tensor-model nodes; the base for rebuilding row 4 inside the gap	torch / CUDA / core change
6	apache/hadoop:3.4.1 image	docker save .tar.gz	1.19 GB	HDFS nodes (unless an HDFS cluster exists)	rarely
7	Model weights: gliner-multi, nllb-200-distilled-600M, hf_cache	folders -> HDFS /models/weights or the master's shared folder	4.80 GB (gliner 2.31 GB)	HDFS / master once; node caches fill themselves	new model version
8	Offline build bundles: wheels/ner_translate (pip wheelhouse), debs/ner_translate (50 .deb files)	folders	wheelhouse several GB (includes a transitively pulled torch / CUDA wheel set); debs 23 MB	the build host inside the gap	NER requirements / apt list change
9	Platform code: the repository without weights, wheels, debs	tar.gz	a few MB	every node (mounted) or the build host	every code change
10	YOLO weights yolov8n.pt / yolov8s.pt - only if the YOLO models are used	files next to the code or at weights_path	6 MB / 22 MB	tensor-model nodes	rarely
11	Checksums (SHA256SUMS) + AIRGAP_BILL_OF_MATERIALS.csv	text files	KB	the receiving side	every transfer
12	Optional: Triton image, the Django backend's Python packages (Django >= 4.2, Celery)	image / wheelhouse	-	if used	-

Image sizes are the images' content size (docker image inspect; close to docker save output before gzip). docker images on the laptop reports the unpacked size on disk, which is larger: 27.7 / 2.2 / 28.1 / 9.3 / 3.3 GB. ner-translate-worker is built on multi-model-inference, so saving both in **one** docker save stores the shared layers once.

Two gaps in what the repository ships today. (1) wheels/ner_translate is empty in this checkout: the wheelhouse the images were built from lives on the build host only. Archive it with every release, or the worker image cannot be rebuilt inside the gap. (2) No image contains the YOLO weights. Offline, yolov8_nano / yolov8_small try to download yolov8n.pt / yolov8s.pt, fail, and **silently** fall back to a stand-in CNN that returns meaningless boxes. Carry the .pt files and pass weights_path, or leave the YOLO models out.

2. What each node needs

node role	images	host software	data / models	notes
Spark master + driver (service mode)	spark-lean	Docker	shared data mount	the driver never imports torch
Spark worker (service mode)	spark-lean	Docker	shared data mount	executors only POST file paths
GPU model server (kitchen)	ner-translate-server	Docker, NVIDIA driver, Container Toolkit	shared data mount; model cache disk >= 5 GB	fetches models from HDFS on first start
Spark master / worker (cluster mode)	ner-translate-worker	Docker (+ NVIDIA stack on GPU workers)	shared data; model cache per node	every executor loads the models
tensor-model cluster	multi-model-inference	Docker (+ NVIDIA stack on GPU workers)	optional YOLO weights	torchvision weights are baked in
HDFS namenode / datanodes	apache/hadoop:3.4.1	Docker	the model weights (put once)	or use an existing HDFS
build host inside the gap (optional)	multi-model-inference (base)	Docker with BuildKit	wheelhouse, .deb bundle, code	rebuilds ner-translate-worker offline
backend (Django gateway)	none	Python >= 3.10, Django >= 4.2, docker CLI or ssh client	shared data mount	see integrations/django/README.md

3. Platform versions per image

	multi-model-inference	spark-lean	ner-translate-worker	ner-translate-server	apache/hadoop
OS	Ubuntu 22.04.5 LTS	Ubuntu 22.04.5 LTS	Ubuntu 22.04.5 LTS	Debian GNU/Linux 12 (bookworm)	CentOS Linux 7 (Core)
Python	Python 3.11.15	Python 3.11.15	Python 3.11.15	Python 3.11.16	-
Java	OpenJDK "17.0.20" 2026-07-21	OpenJDK "17.0.20" 2026-07-21	OpenJDK "17.0.20" 2026-07-21	-	OpenJDK "1.8.0_212"
Spark	Spark 3.5.1 (Hadoop 3.3.4 client)	Spark 3.5.1 (Hadoop 3.3.4 client)	Spark 3.5.1 (Hadoop 3.3.4 client)	-	-
Hadoop	-	-	-	-	Hadoop 3.4.1
torch / CUDA / cuDNN	2.6.0+cu126 / 12.6 / 9.5.1	-	2.6.0+cu126 / 12.6 / 9.5.1	2.6.0+cu126 / 12.6 / 9.5.1	-
tesseract	-	-	tesseract 4.1.1	tesseract 5.3.0	-
OCR languages	-	-	ben eng guj hin kan mal mar osd pan tam tel urd	ben eng guj hin kan mal mar osd pan tam tel urd	-
pip packages / dpkg packages	82 / 281	10 / 174	122 / 328	79 / 176	0 / 0
image ID (short)	986e9e56cddf	29a195a86a9f	46a0d21a8364	820922c1fb83	69ffa97339af
built	2026-09-16	2026-09-16	2026-09-17	2026-09-16	2024-11-06

GPU code in the torch build of all three torch images: sm_50, sm_60, sm_70, sm_75, sm_80, sm_86, sm_90 (Maxwell to Hopper; the laptop's GTX 1650 is sm_75, a T4 sm_75, an A100 sm_80, an H100 sm_90). There is no sm_100 / sm_120: Blackwell GPUs such as the RTX 50 series need a torch build for CUDA 12.8 or later. glibc: 2.35 in the Ubuntu images, 2.36 in the kitchen (Debian 12). All images are linux/amd64.

Build host used for this inventory: Docker client 29.5.3 / server 29.5.3 (linux), Compose 5.1.4, GPU / driver NVIDIA GeForce GTX 1650, 610.62, 4096 MiB.

4. Python packages and why they are pinned

package	multi-model-inference	spark-lean	ner-translate-worker	ner-translate-server	why this version / note
pyspark	3.5.1	3.5.1	3.5.1	-	must equal the Spark version (3.5.1) on every node
py4j	0.10.9.7	0.10.9.7	0.10.9.7	-	ships with pyspark 3.5.1
torch	2.6.0+cu126	-	2.6.0+cu126	2.6.0+cu126	cu126 wheel from download.pytorch.org, installed by the Dockerfile (not requirements.txt)
torchvision	0.21.0+cu126	-	0.21.0+cu126	0.21.0+cu126	installed together with torch; pinning it in requirements.txt once downgraded torch to 2.2
numpy	2.1.3	-	2.1.3	2.4.6	2.x; core requirements pin 2.1.3 - the kitchen resolves it unpinned (see section 10)
pandas	2.2.3	-	2.2.3	-	core, for Spark / Arrow
pyarrow	18.1.0	-	18.1.0	18.1.0	18.1.0; imported before torch in the NER module (OpenMP/MKL clash)
transformers	-	-	4.45.2	4.45.2	4.45.2: 4.57.x breaks loading the GLiNER tokenizer (list vs dict special tokens)
tokenizers	-	-	0.20.3	0.20.3	transitive, follows transformers
gliner	-	-	0.2.13	0.2.13	0.2.13, developed against transformers 4.45
sentencepiece	-	-	0.2.2	0.2.2	0.2.2: the first with a manylinux_2_28 wheel
tiktoken	-	-	0.12.0	0.12.0	0.12.0: same wheel-tag reason; tokenizer conversion needs it or protobuf
protobuf	-	-	5.29.3	5.29.3	5.29.3: needed to convert GLiNER's tokenizer
onnxruntime	-	-	1.30.0	1.30.0	1.30.0: older (1.16) crashes with numpy 2 (ABI)
huggingface-hub	-	-	0.36.2	0.36.2	transitive (gliner / transformers); hf-xet forced into the wheelhouse
safetensors	-	-	0.8.0	0.8.0	transitive; GLiNER loads model.safetensors
py3langid	-	-	0.4.0	0.4.0	0.4.0, needs numpy >= 2
pillow	12.3.0	-	11.0.0	11.0.0	11.0.0 pinned by NER (the base resolves 12.x, the NER install downgrades it)
pytesseract	-	-	0.3.13	0.3.13	0.3.13 - calls the tesseract binary
pypdf	-	-	5.1.0	5.1.0	5.1.0
pdf2image	-	-	1.17.0	1.17.0	1.17.0 - needs poppler-utils
fastapi	-	-	0.121.2	0.121.2	0.121.2 - the kitchen
uvicorn	-	-	0.38.0	0.38.0	0.38.0 - the kitchen
requests	2.34.2	2.34.2	2.34.2	2.34.2	the lean executors' HTTP client
ultralytics	8.3.40	-	8.3.40	-	8.3.40 - YOLO; pulls opencv-python and scipy unpinned
opencv-python	5.0.0.93	-	5.0.0.93	-	transitive via ultralytics (unpinned)
nvidia-cudnn-cu12	9.5.1.17	-	9.5.1.17	9.5.1.17	CUDA runtime libraries arrive as pip packages with torch
nvidia-cublas-cu12	12.6.4.1	-	12.6.4.1	12.6.4.1	same
triton	3.2.0	-	3.2.0	3.2.0	OpenAI Triton compiler used by torch, not the Triton server
boto3	1.35.99	-	1.35.99	-	S3 upload of results on AWS only

Where the pins live: requirements.txt (core, installed in the base image), models/pipelines/ner_translate/requirements.txt (the pipeline's own, installed in both NER images), and the Dockerfiles (torch, torchvision, pyspark and requests for the lean image). The full list of every package in every image is Appendix A.

5. System packages (.deb bundle)

debs/ner_translate holds 50 .deb files (23 MB): tesseract-ocr 4.1.1, 12 language data packages (the 11 OCR languages + osd), poppler-utils 22.02 and their full dependency closure, downloaded **from inside multi-model-inference:latest** so each version matches what is already installed in that image (a fresh ubuntu:22.04 once resolved a newer libgomp1 than the image's gcc-12-base and dpkg refused to configure it). Appendix B lists every package and version.

The kitchen image does not use this bundle: it installs tesseract 5.3.0 and poppler from Debian 12's apt repositories at build time (see section 10).

6. Model weights

file	size	SHA-256 (first 16)	needed?
gliner-multi/model.safetensors	1,155.8 MB	2100142f31627531	yes
gliner-multi/pytorch_model.bin	1,155.9 MB	580d8b061bcb6596	no - GLiNER loads model.safetensors
nllb-200-distilled-600M/pytorch_model.bin	2,460.5 MB	c266c2cfd19758b6	yes
nllb-200-distilled-600M/sentencepiece.bpe.model	4.9 MB	14bb8dfb35c0ffde	yes
nllb-200-distilled-600M/tokenizer.json	17.3 MB	e316b82de11d0f95	yes
hf_cache/mdebarta blobs/13c8d666d62a7bc4ac8f040aab68e942c861f93303156c	4.3 MB	13c8d666d62a7bc4	yes

Plus 38 small files (configs, tokenizer files, the Hugging Face cache's refs and symlinks). Full checksums: Appendix C.

model	source (Hugging Face)	used by	licence (check the model card)
gliner-multi	urchade/gliner_multi-v2.1	entity extraction	Apache-2.0 per its model card
nllb-200-distilled-600M	facebook/nllb-200-distilled-600M	translation to English	CC-BY-NC-4.0 - non-commercial; clear it for your use
hf_cache	microsoft/mdebarta-v3-base, snapshot a0484667... (config + tokenizer only)	GLiNER's backbone definition	MIT
torchvision checkpoints (baked in the image)	resnet18-f37072fd, mobilenet_v3_small-047dcff4, efficientnet_b0_rwightman-7f5810bc	image models	BSD-3 (torchvision)
yolov8n.pt / yolov8s.pt (not shipped)	Ultralytics	YOLO models	AGPL-3.0 (Ultralytics) - check before shipping

Where they go: once into HDFS (hdfs dfs -put gliner-multi nllb-200-distilled-600M hf_cache /models/weights/) or the master's shared folder; nodes fetch them into their local cache on first use (models/model_store.py). A new model version goes into a new HDFS folder and MODEL_STORE_URI points at it - old and new caches coexist.

7. Version compatibility rules

must match	rule	what breaks otherwise
NVIDIA driver vs CUDA runtime (12.6)	>= 525.60.13 works through CUDA 12 minor-version compatibility; >= 560.28 is the native 12.6 requirement (recommended)	torch.cuda.is_available() False / CUDA init errors
GPU compute capability	sm_50-sm_90 (section 3)	'no kernel image is available for execution on the device'
Python minor version, driver = executors	3.11 everywhere (the images guarantee it)	PYTHON_VERSION_MISMATCH - seen on a Windows host with 3.12 + 3.10
pyspark = Spark	3.5.1 on the driver and in every executor image	Py4J / protocol errors
Java for Spark	17 (images); Spark 3.5 supports 8 / 11 / 17	JVM start failures
torch = torchvision pair	2.6.0 + 0.21.0, both cu126	import errors / silent torch downgrade
transformers vs gliner	4.45.2 with gliner 0.2.13	GLiNER tokenizer load fails ('list' object has no attribute 'keys')
numpy 2 ABI	onnxruntime >= 1.17 (1.30.0), pyarrow 18, py3langid 0.4	'_ARRAY_API not found' at import
wheel platform tags	manylinux2014 / manylinux_2_28 wheels need glibc >= 2.17 / 2.28 (images have 2.35 / 2.36)	pip: no matching distribution (offline)
driver and all workers, same wheels	identical images, or identical wheels-hotfix contents on every node	different results or unpickling errors between driver and executors
kitchen vs worker (NER)	same model files; OCR differs today (tesseract 5.3 vs 4.1)	different OCR text for scanned documents between modes

8. Dependency management

8.1 Who owns which dependency

28. Docker images - what is built from what

Three Dockerfiles, four images; arrows = FROM / build target. Sizes as listed by docker images on the laptop

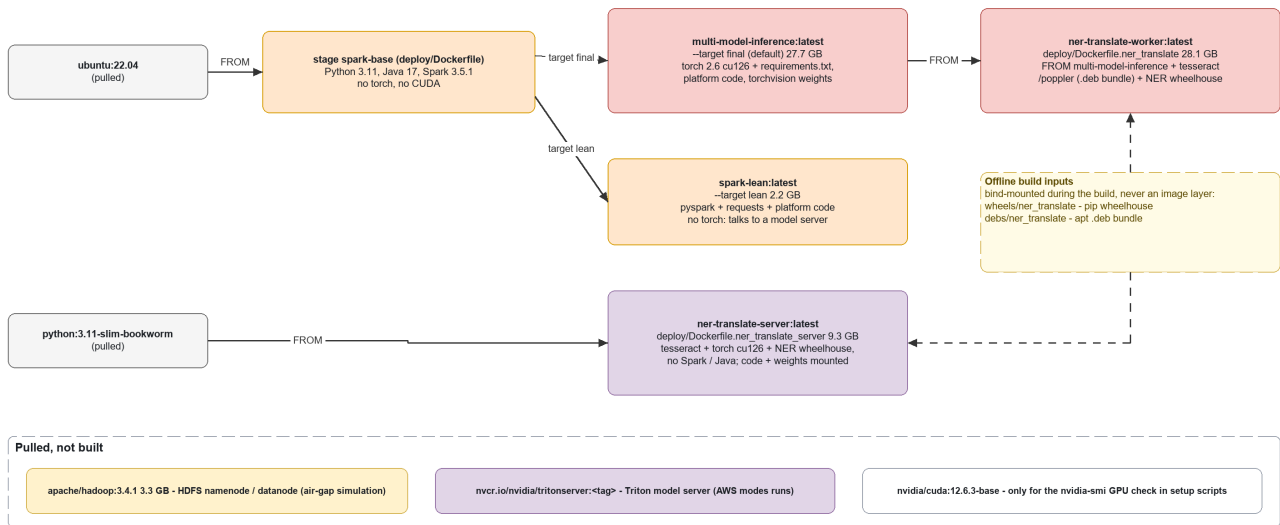


Figure 1 - image layering: each layer owns its own dependencies

- **requirements.txt** - the core set for every tensor model (pyspark, numpy, pandas, pyarrow, ultralytics, ...). Installed only in the base image. Pipelines never add to it: one pipeline's dependency once silently changed a version every other model relied on.
- **models/pipelines/<name>/requirements.txt** - one pipeline's own packages, installed in that pipeline's images only. It must not re-pin what the base provides (torch, numpy, pandas), except where the kitchen, which has no base, needs it listed.
- **The Dockerfiles** - torch / torchvision (from the CUDA wheel index), Java, Spark, OS packages.
- **Model weights** - never in an image (except the three small torchvision checkpoints): they live in the model store.

8.2 Building offline-capable images

step	command / mechanism	why
1 wheelhouse	bash deploy/scripts/build_ner_translate_wheelhouse.sh -> wheels/ner_translate	pip download for the container (--platform manylinux2014_x86_64 and manylinux_2_28_x86_64, --python-version 311, --only-binary), sdist-only odfpy / ebooklib downloaded separately, their real deps (defusedxml, lxml, six) and hf-xet forced in - the environment markers are evaluated for the build host, not the container
2 .deb bundle	bash deploy/scripts/build_ner_translate_debs.sh -> debs/ner_translate	apt --download-only inside multi-model-inference itself, so versions match the installed base
3 build	docker build -t ner-translate-worker:latest -f deploy/Dockerfile.ner_translate .	pip install --no-index --find-links (fails loudly if a pin is missing) and dpkg -i twice; both bundles bind-mounted, never COPY'd, so they are not shipped as image layers
4 inventory	python docs/airgap/collect_inventory.py	records what the images really contain (this document)

8.3 What can be rebuilt inside the gap

image	rebuild inside the gap?	what it needs
ner-translate-worker	yes	multi-model-inference already loaded + wheels/ner_translate + debs/ner_translate + code
multi-model-inference, spark-lean	no	ubuntu:22.04, the deadsnakes PPA, apt, the Spark tarball from archive.apache.org, download.pytorch.org and PyPI - rebuild outside and carry the image
ner-translate-server	no	Debian apt mirrors and download.pytorch.org at build time; no hotfix hook - carry a new image for any change
apache/hadoop	no (pulled)	carry the image

8.4 Updating inside the gap - what to carry for each kind of change

change	carry	then
platform code (inference/, models/, submit_*.py)	code tar.gz (MB)	extract to the mounted folder on every node; restart containers
new model version	the model folder	hdfs dfs -put to a new folder; point MODEL_STORE_URI at it
one NER Python package (urgent)	the .whl files	wheels-hotfix/ner_translate on every node, restart (worker image only)
NER Python packages (planned)	new wheelhouse	rebuild ner-translate-worker offline (8.2 step 3)
OCR / system packages for the worker	new .deb bundle	rebuild ner-translate-worker offline
kitchen dependencies	new ner-translate-server image	docker load on the GPU node
torch / CUDA / Python / Java / Spark / core packages	new images	docker load on every node; rebuild the worker FROM the new base

wheels-hotfix: deploy/apply_wheels_hotfix.sh runs at start in the ner-translate-worker containers and installs any .whl in the /wheels-hotfix mount. It is for one urgent fix without a rebuild. The same wheels must be on every node, because a driver and executors with different package versions is a real failure mode. Fold the fix into the wheelhouse at the next release.

8.5 Recommended: lock the transitive versions

Direct dependencies are pinned, but their dependencies are resolved at build time and drift between builds. This inventory shows it: numpy is 2.1.3 in the worker but 2.4.6 in the kitchen; ultralytics pulls opencv-python and scipy unpinned. To make a rebuild reproduce exactly what was tested, freeze each tested image and build with the result as a constraints file:

```
docker run --rm --entrypoint python3 ner-translate-server:latest -m pip freeze \
  > constraints/ner-translate-server.txt
docker run --rm --entrypoint python3 multi-model-inference:latest -m pip freeze \
  > constraints/multi-model-inference.txt
# in the Dockerfiles / wheelhouse script: pip install ... -c constraints/<image>.txt
# and pin numpy==2.1.3 in models/pipelines/ner_translate/requirements.txt so both NER images agree
```

9. Packaging, transfer and verification

```
# ---- outside (connected build host) ----
python docs/airgap/collect_inventory.py                                # inventory.json + BILL_OF_MATERIALS.csv
# one save for both torch images: their shared layers are stored once
docker save multi-model-inference:latest ner-translate-worker:latest | gzip > spark-torch-images.tar.gz
docker save spark-lean:latest | gzip > spark-lean.tar.gz
docker save ner-translate-server:latest | gzip > ner-translate-server.tar.gz
docker save apache/hadoop:3.4.1 | gzip > hadoop.tar.gz
tar czf models-weights.tar.gz -C <weights dir> gliner-multi nllb-200-distilled-600M hf_cache
tar czf build-bundles.tar.gz wheels/ner_translate debs/ner_translate
tar czf platform-code.tar.gz --exclude=models/weights --exclude=wheels --exclude=debs \
  --exclude=results --exclude=.git .
sha256sum *.tar.gz docs/airgap/AIRGAP_BILL_OF_MATERIALS.csv > SHA256SUMS

# ---- inside ----
sha256sum -c SHA256SUMS                                              # every file intact
for f in *images*.tar.gz spark-lean.tar.gz ner-translate-server.tar.gz hadoop.tar.gz; do docker load -i $f; done
docker image inspect --format '{{.Id}}' spark-lean:latest           # = the ID in the bill of materials
python docs/airgap/collect_inventory.py --no-hash                   # diff inventory.json against the one carried in
```

check inside the gap	command	expected
GPU reachable from containers	docker run --rm --gpus all --entrypoint nvidia-smi ner-translate-server:latest	the GPU and driver listed
torch + CUDA	docker run --rm --gpus all --entrypoint python3 ner-translate-server:latest -c "import torch;print(torch.cuda.is_available())"	True
NER packages	... -c "import gliner, transformers;print(gliner.__version__, transformers.__version__)"	0.2.13 4.45.2
OCR languages	docker run --rm --entrypoint tesseract ner-translate-server:latest --list-langs	12 entries (11 + osd)
Spark	docker run --rm spark-lean:latest cat /opt/spark/RELEASE	Spark 3.5.1
models in HDFS	hdfs dfs -du -s -h /models/weights/*	sizes as in section 6
end to end	.\deploy\airgap_sim_tests.ps1 (simulation) or one submit_pipeline_job.py run	all tests PASS / 6 documents processed

10. Risks found in this inventory

risk	evidence	recommendation
NER images disagree on numpy	worker 2.1.3, kitchen 2.4.6	pin numpy==2.1.3 in the NER requirements; constraints files (8.5)
OCR engine differs between modes	worker tesseract 4.1.1, kitchen 5.3.0	build the kitchen from the same Ubuntu base + .deb bundle, or accept and document the difference
kitchen image cannot be rebuilt offline, has no hotfix hook	Dockerfile.ner_translate_server: apt + download.pytorch.org at build	base it on a local image + wheelhouse like the worker, or plan to carry a new image per change
wheelhouse not archived with the code	wheels/ner_translate empty in this checkout	archive wheels/ + deps/ with each release
YOLO weights missing	no image contains yolov8n.pt / yolov8s.pt; silent fallback CNN	carry the .pt files or exclude the YOLO models
unpinned transitive dependencies	opencv-python 5.0.0.93, scipy 1.17.1 via ultralytics	constraints files (8.5)
plain docker build gives the wrong image	Dockerfile default target is lean	build with --target final (see DOCKER_CONTAINERS_AND_COMPOSE.pdf)
1.2 GB of unused weights	gliner-multi/pytorch_model.bin never read	leave it out of HDFS and the transfer
licences	NLLB-200 is CC-BY-NC-4.0; Ultralytics is AGPL-3.0	clear with your legal / security team before deployment
no Blackwell support	torch arch list ends at sm_90	choose sm_90-or-older GPUs, or rebuild with a CUDA 12.8+ torch

Appendix A - every Python package per image

multi-model-inference (82 packages)

package==version		
blinker==1.4	networkx==3.6.1	pyspark==3.5.1
boto3==1.35.99	numpy==2.1.3	python-apt==2.4.0+ubuntu4.1
botocore==1.35.99	nvidia-cublas-cu12==12.6.4.1	python-dateutil==2.9.0.post0
certifi==2026.7.22	nvidia-cuda-cupti-cu12==12.6.80	pytz==2026.3.post1
charset-normalizer==3.5.1	nvidia-cuda-nvrtc-cu12==12.6.77	pyyaml==6.0.3
contourpy==1.3.3	nvidia-cuda-runtime-cu12==12.6.77	requests==2.34.2
cryptography==3.4.8	nvidia-cudnn-cu12==9.5.1.17	s3transfer==0.10.4
cycler==0.12.1	nvidia-cufft-cu12==11.3.0.4	scipy==1.17.1
dbus-python==1.2.18	nvidia-curand-cu12==10.3.7.77	seaborn==0.13.2
distro==1.7.0	nvidia-cusolver-cu12==11.7.1.2	secretstorage==3.3.1
filelock==3.32.3	nvidia-cuspars-cu12==12.5.4.2	setuptools==59.6.0
fonttools==4.65.0	nvidia-cusparselt-cu12==0.6.3	six==1.16.0
fsspec==2026.7.0	nvidia-nccl-cu12==2.21.5	sympy==1.13.1
httplib2==0.20.2	nvidia-nvjitlink-cu12==12.6.85	tabulate==0.9.0
idna==3.19	nvidia-nvtx-cu12==12.6.77	torch==2.6.0+cu126
importlib-metadata==4.6.4	oauthlib==3.2.0	torchvision==0.21.0+cu126
jeepney==0.7.1	opencv-python==5.0.0.93	tqdm==4.70.1
jinja2==3.1.6	packaging==26.3	triton==3.2.0
jmespath==1.1.0	pandas==2.2.3	typing-extensions==4.16.0
keyring==23.5.0	pillow==12.3.0	tzdata==2026.4
kiwisolver==1.5.1	pip==22.0.2	ultralitics==8.3.40
launchpadlib==1.10.16	psutil==7.2.2	ultralitics-thop==2.1.6
lazr.restfulclient==0.14.4	py-cpuinfo==9.0.0	urllib3==2.8.0
lazr.uri==1.0.6	py4j==0.10.9.7	wadllib==1.3.6
markupsafe==3.0.3	pyarrow==18.1.0	wheel==0.37.1
matplotlib==3.9.2	pygobject==3.42.1	zipp==1.0.0
more-itertools==8.10.0	pyjwt==2.3.0	
mpmath==1.3.0	pyparsing==2.4.7	

spark-lean (10 packages)

package==version		
certifi==2026.7.22	py4j==0.10.9.7	urllib3==2.8.0
charset-normalizer==3.5.1	pyspark==3.5.1	wheel==0.37.1
idna==3.19	requests==2.34.2	
pip==22.0.2	setuptools==59.6.0	

ner-translate-worker - packages added or changed on top of multi-model-inference

package==version		
annotated-doc==0.0.5	lxml==6.1.3	regex==2026.9.10
annotated-types==0.8.0	odfpy==1.4.1	safetensors==0.8.0
anyio==4.15.1	onnxruntime==1.30.0	sentencepiece==0.2.2
beautifulsoup4==4.12.3	openpyxl==3.1.5	soupsieve==2.9.2
click==8.5.0	pdf2image==1.17.0	starlette==0.49.3
defusedxml==0.7.1	pillow==11.0.0	stripptf==0.0.28
ebooklib==0.18	protobuf==5.29.3	tiktoken==0.12.0
et-xmlfile==2.0.0	py3langid==0.4.0	tokenizers==0.20.3
fastapi==0.121.2	pydantic==2.13.5	transformers==4.45.2
flatbuffers==25.12.19	pydantic-core==2.46.5	typing-inspection==0.4.4
gliner==0.2.13	pypdf==5.1.0	uvicorn==0.38.0
h11==0.16.0	pytesseract==0.3.13	xlrd==2.0.1
hf-xet==1.6.0	python-docx==1.1.2	xlswriter==3.2.9
huggingface-hub==0.36.2	python-pptx==1.0.2	

ner-translate-server (79 packages)

package==version		
annotated-doc==0.0.5	nvidia-cuda-nvrtc-cu12==12.6.77	pyyaml==6.0.3
annotated-types==0.8.0	nvidia-cuda-runtime-cu12==12.6.77	regex==2026.9.10
anyio==4.15.1	nvidia-cudnn-cu12==9.5.1.17	requests==2.34.2
beautifulsoup4==4.12.3	nvidia-cufft-cu12==11.3.0.4	safetensors==0.8.0
certifi==2026.7.22	nvidia-curand-cu12==10.3.7.77	sentencepiece==0.2.2
charset-normalizer==3.5.1	nvidia-cusolver-cu12==11.7.1.2	setuptools==79.0.1
click==8.5.0	nvidia-cusparse-cu12==12.5.4.2	six==1.17.0
defusedxml==0.7.1	nvidia-cusparselt-cu12==0.6.3	soupsieve==2.9.2
ebooklib==0.18	nvidia-nccl-cu12==2.21.5	starlette==0.49.3
et-xmlfile==2.0.0	nvidia-nvjitlink-cu12==12.6.85	stripptf==0.0.28
fastapi==0.121.2	nvidia-nvtx-cu12==12.6.77	sympy==1.13.1
filelock==3.32.3	odfpy==1.4.1	tiktoken==0.12.0
flatbuffers==25.12.19	onnxruntime==1.30.0	tokenizers==0.20.3
fsspec==2026.7.0	openpyxl==3.1.5	torch==2.6.0+cu126
gliner==0.2.13	packaging==26.3	torchvision==0.21.0+cu126
h11==0.16.0	pdf2image==1.17.0	tqdm==4.70.1
hf-xet==1.6.0	pillow==11.0.0	transformers==4.45.2
huggingface-hub==0.36.2	pip==24.0	triton==3.2.0
idna==3.19	protobuf==5.29.3	typing-extensions==4.16.0
jinja2==3.1.6	py3langid==0.4.0	typing-inspection==0.4.4
lxml==6.1.3	pyarrow==18.1.0	urllib3==2.8.0
markupsafe==3.0.3	pydantic==2.13.5	uvicorn==0.38.0
mpmath==1.3.0	pydantic-core==2.46.5	wheel==0.46.3
networkx==3.6.1	pypdf==5.1.0	xlrd==2.0.1
numpy==2.4.6	pytesseract==0.3.13	xlswriter==3.2.9
nvidia-cublas-cu12==12.6.4.1	python-docx==1.1.2	
nvidia-cuda-cupti-cu12==12.6.80	python-pptx==1.0.2	

Appendix B - .deb bundle (debs/ner_translate)

package	version	package	version
fontconfig	2.13.1-4.2ubuntu5	libwebp7	1.2.2-2ubuntu0.22.04.2
gcc-12-base	12.3.0-1ubuntu1~22.04.3	libwebpmux3	1.2.2-2ubuntu0.22.04.2
libarchive13	3.6.0-1ubuntu1.8	libx11-6	2:1.7.5-1ubuntu0.3
libbsd0	0.11.5-1	libx11-data	2:1.7.5-1ubuntu0.3
libcairo2	1.16.0-5ubuntu2.1	libxau6	1:1.0.9-1build5
libdatrie1	0.2.13-2	libxcb-render0	1.14-3ubuntu3
libdeflate0	1.10-2	libxcb-shm0	1.14-3ubuntu3
libfribidi0	1.0.8-2ubuntu3.1	libxcb1	1.14-3ubuntu3
libgcc-s1	12.3.0-1ubuntu1~22.04.3	libxdmcp6	1:1.1.3-0ubuntu5
libgif7	5.1.9-2ubuntu0.3	libxext6	2:1.3.4-1build1
libgomp1	12.3.0-1ubuntu1~22.04.3	libxrender1	1:0.9.10-1build4
libjbig0	2.1-3.1ubuntu0.22.04.1	poppler-utils	22.02.0-2ubuntu0.13
liblpt5	1.82.0-3build1	tesseract-ocr-ben	1:4.00~git30-7274cfa-1.1
libmd0	1.0.4-1build1	tesseract-ocr-eng	1:4.00~git30-7274cfa-1.1
libopenjp2-7	2.4.0-6ubuntu0.5	tesseract-ocr-guj	1:4.00~git30-7274cfa-1.1
libpango-1.0-0	1.50.6+ds-2ubuntu1	tesseract-ocr-hin	1:4.00~git30-7274cfa-1.1
libpangocairo-1.0-0	1.50.6+ds-2ubuntu1	tesseract-ocr-kan	1:4.00~git30-7274cfa-1.1
libpangoft2-1.0-0	1.50.6+ds-2ubuntu1	tesseract-ocr-mal	1:4.00~git30-7274cfa-1.1
libpixman-1-0	0.40.0-1ubuntu0.22.04.1	tesseract-ocr-mar	1:4.00~git30-7274cfa-1.1
libpoppler118	22.02.0-2ubuntu0.13	tesseract-ocr-osd	1:4.00~git30-7274cfa-1.1
libstdc++6	12.3.0-1ubuntu1~22.04.3	tesseract-ocr-pan	1:4.00~git30-7274cfa-1.1
libtesseract4	4.1.1-2.1build1	tesseract-ocr-tam	1:4.00~git30-7274cfa-1.1
libthai-data	0.1.29-1build1	tesseract-ocr-tel	1:4.00~git30-7274cfa-1.1
libthai0	0.1.29-1build1	tesseract-ocr-urd	1:4.00~git30-7274cfa-1.1
libtiff5	4.3.0-6ubuntu0.13	tesseract-ocr	4.1.1-2.1build1

Appendix C - model file checksums

file	bytes	SHA-256
gliner-multi/.gitattributes	1,519	11ad7efa24975ee4b0c3c3a38ed18737f0658a5f75a0a96787b576a78a023361
gliner-multi/README.md	4,770	820125f2ea897716cc38645d94f73f390ec9dfd392980c436be1d25c2e106b55
gliner-multi/gliner_config.json	477	e25f61d91620df84aae8076811ee592e926e94d341b82e7bc1be359718f83017
gliner-multi/model.safetensors	1,155,830,112	2100142f31627531497850659dcb3821c99d5e71c08a8e01a98e4b11ef32a199
gliner-multi/pytorch_model.bin	1,155,900,362	580d8b061bcb6596bd79be7f29736cea84fdfe0679fc3ae9c14c08d2e974fad
gliner-multi/.cache/huggingface/.gitignore	1	684888c0ebb17f374298b65ee2807526c066094c701bcc7ebbe1c1095f494fc1
gliner-multi/.cache/huggingface/download/.gitattributes.metadata	104	ab6cb87d27f6f888240a7db77bf11881be8fe69629fceb5e4883264e7718435
gliner-multi/.cache/huggingface/download/README.md.metadata	104	35566fba036c194e1f37bd0f57a9e2a0b9f6a91bc5a6b9786e56fafa96bdf491
gliner-multi/.cache/huggingface/download/gliner_confii.metadata	104	6bc1b478c3eaf354b849f23e748efb75f38d3a2fc04e43945c9dc5d3e8ac9e4e
gliner-multi/.cache/huggingface/download/model.safete..metadata	128	34b21411ec40f9751e865f9fed22d21ef567ae3c5fd8208637a08a5b3c40b5c2
gliner-multi/.cache/huggingface/download/pytorch_model..metadata	127	83fb79860cd8f4ee4e897bfef04b96277612bbd1c767b0e9e1c121f0a80906a4
nllb-200-distilled-600M/.gitattributes	1,284	2ba66c0e2f069c4a2c7c2edbfdfad547a21657118f74c6f4667a41e87b6b18be
nllb-200-distilled-600M/README.md	7,667	0964c46d35be00825c73b3a9ca212043b4b4e4f38e18e88c22e4490d31f9f1a3
nllb-200-distilled-600M/config.json	846	f9b4081d3d108e9d06532ea8d92b932c04ed3a15ff2f87e6982f8e14db51fbc5
nllb-200-distilled-600M/generation_config.json	189	0bb604cbb8392649176935a447ce994128f170a6908c135a17d0e3dfb8113cb2
nllb-200-distilled-600M/pytorch_model.bin	2,460,457,927	c266c2cfd19758b6d09c1fc31ecdf1e485509035f6b51dfe84f1ada83eefcc42
nllb-200-distilled-600M/sentencepiece.bpe.model	4,852,054	14bb8dfb35c0ffdea7bc01e56cea38b9e3d5efcdcb9c251d6b40538e1aab555a
nllb-200-distilled-600M/special_tokens_map.json	3,548	992bd4ed610d644d6823081937bcc91bb8878dd556cea4ae5327f2480361330e
nllb-200-distilled-600M/tokenizer.json	17,331,176	e316b82de11d0f951f370943b3c438311629547285129b0b81dadabd01bca665
nllb-200-distilled-600M/tokenizer_config.json	564	d1aa8c3697d3e35674f97b5b7e9c99d22b010f528e80140257d97316be90d044
nllb-200-distilled-600M/.cache/huggingface/.gitignore	1	684888c0ebb17f374298b65ee2807526c066094c701bcc7ebbe1c1095f494fc1
nllb-200-distilled-600M/.cache/huggingface/download/.gitattributes.metadata	104	8f1045dbf0dd6461b73a547ca317037e8e39f2f622378dfd1ae3d31a8604187a
nllb-200-distilled-600M/.cache/huggingface/download/README.md.metadata	103	615ced55e9830b24919a9ebb01f2fb341610b02a0eccc12452a4fa359f7315aa
nllb-200-distilled-600M/.cache/huggingface/download/config.json.metadata	104	adf2c6374196bae8fec43d0d8c239c9a9d04eeaac48d7a09e604fd92dcb8b0c7
nllb-200-distilled-600M/.cache/huggingface/download/generation_c..metadata	104	df44e6e192766003c0d5ab93c638b8b84f1b6a2fb88e0f475516b63bca7c9b87
nllb-200-distilled-600M/.cache/huggingface/download/pytorch_model..metadata	128	d45cc9182bbb2da7cb538c6cc9519cfd694aef17067cf340ccae4d11029c195
nllb-200-distilled-600M/.cache/huggingface/download/sentencepiec..metadata	128	2cf67938134ec9c4deac5eb62994259e4d3de7fb8e92d0ed9f71be2e2717ef42
nllb-200-distilled-600M/.cache/huggingface/download/special_toke..metadata	104	04b2ad04d4d54851cc823a60f2fff90929626446ea7f11eff1a2e828a2eeaf79
nllb-200-distilled-600M/.cache/huggingface/download/tokenizer.json.metadata	126	3332e5e4a43808bcbf08ca6153ffc6fd64975c18c5674fd98688ffccd664a9cd
nllb-200-distilled-600M/.cache/huggingface/download/tokenizer_co..metadata	103	40966b0614d5a7a6b4e0a14c50d38b4839aa77d276cded8ffa88101863c96400
hf_cache/version.txt	1	6b86b273ff34fce19d6b804eff5a3f5747ada4eaa22f1d49c01e52ddb7875b4b
hf_cache/.locks/models--micr...v3-base/13c8d666d62a..6e3.lock	0	empty marker file (lock / no-exist entry)
hf_cache/.locks/models--micr...v3-base/27b905d788d2..ea7.lock	0	empty marker file (lock / no-exist entry)
hf_cache/.locks/models--micr...v3-base/acfd94e399c5..7a6.lock	0	empty marker file (lock / no-exist entry)
hf_cache/<mdeberta>/.no_exist/a0484667../added_tokens.json	0	empty marker file (lock / no-exist entry)
hf_cache/<mdeberta>/.no_exist/a0484667../special_tokens_map.json	0	empty marker file (lock / no-exist entry)

file	bytes	SHA-256
hf_cache/<mdeberta>/no_exist/a0484667../tokenizer.json	0	empty marker file (lock / no-exist entry)
hf_cache/<mdeberta>/blobs/13c8d666d62a..5d86d6e3	4,305,025	13c8d666d62a7bc4ac8f040aab68e942c861f93303156cc28f5c7e885d86d6e3
hf_cache/<mdeberta>/blobs/27b905d788d2..23e38ea7	579	bcffcd343dc5efa5ef2d5a58d2b405eed108f01cc45b48d0a907b333ec41801f
hf_cache/<mdeberta>/blobs/acfd94e399c5..111fc7a6	52	3f3978e0c036f2c2588cac34a6047cbb0af0b0dc1814254e291028529805496d
hf_cache/<mdeberta>/refs/main	40	4893177b0d5b48a1f673b82df93f7fac63d83dac14a0c302b44f1f9d83f36e69
hf_cache/<mdeberta>/snapshots/a0484667../config.json	symlink	points to one of the blobs (Hugging Face cache layout)
hf_cache/<mdeberta>/snapshots/a0484667../spm.model	symlink	points to one of the blobs (Hugging Face cache layout)
hf_cache/<mdeberta>/snapshots/a0484667../tokenizer_config.json	symlink	points to one of the blobs (Hugging Face cache layout)

Weights folder hashed: D:\pytorch-spark-inference-platform_20260921\models\weights. Verify on the far side with sha256sum against these values or the bill of materials.